# The Metric Topology

**Def'n:** A metric on a set $X$ is a function $d: X \times X \to \mathbb{R}$ with the following properties:

1) $d(x,y) \geq 0$ for all $x, y \in X$. $d(x,y) = 0 \iff x = y$.

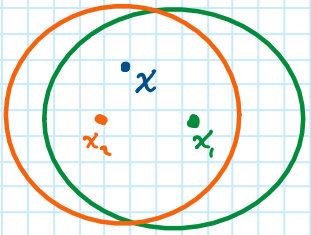
2) $d(x,y) = d(y,x)$ for all $x, y \in X$.

3) $d(x,y) + d(y,z) \geq d(x,z)$ for all $x, y, z \in X$.

**Definition:** If $d$ is a metric on $X$, then the collection of all $\varepsilon$-balls, $B_d(x, \varepsilon)$, for $x \in X$ and $\varepsilon > 0$, is a basis for a topology on $X$ called the metric topology induced by $d$.

**Proof:** for all $x \in X$, there is an $\varepsilon$ ball centered at $x$, thus condition 1 is satisfied.
Suppose $x \in B(x_1, \varepsilon_1) \cap B(x_2, \varepsilon_2)$.

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Then let $\delta = \min\{\varepsilon_1 - d(x, x_1), \varepsilon_2 - d(x, x_2)\}$.
Then $B(x, \delta)$ is a basis element that is in $B(x_1, \varepsilon_1) \cap B(x_2, \varepsilon_2)$ and contains $x$. □

**Def'n:** A set $U$ is open in the metric topology induced by $d$ iff for each $y \in U$, there is a $\delta > 0$ such that $B_d(y, \delta) \subseteq U$.

**Def'n:** $X$ is metrizable if there exists a metric $d$ on $X$ that induces the topology of $X$. A metric space is a metrizable space $X$ with a metric.

**Lemma:** Let $d$ and $d'$ be two metrics on $X$. Let $\tau, \tau'$ be the topologies they induce. Then $\tau'$ is finer than $\tau$ iff for each $x \in X$ and $\varepsilon > 0$, there exists $\delta > 0$ such that $B_{d'}(x, \delta) \subseteq B_d(x, \varepsilon)$.

**Proof:** Suppose $\tau'$ is finer than $\tau$. Then for each basis element of $\tau$, there is a basis element of $\tau'$ contained inside. So let $x \in B_d(x, \varepsilon)$. Then $\exists B'$ in the basis of $\tau'$ s.t. $x \in B' \subseteq B_d(x, \varepsilon)$. By def'n of open in a metric topology, there is a ball $B_{d'}(x, \delta)$ contained in $B'$.

Now suppose $\delta$-$\varepsilon$ condition holds. Then for any basis element $B_d(x, \varepsilon)$, there is a $\delta$ s.t. $B_{d'}(x, \delta) \subseteq B_d(x, \varepsilon)$. Thus $\tau'$ is finer than $\tau$. □

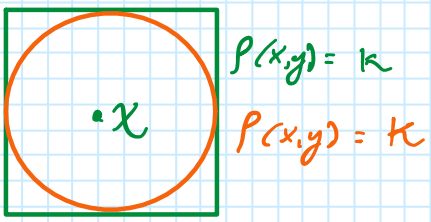
**Def'n:** Let $x \in \mathbb{R}^n$. Define $\|x\|_2 = (x_1^2 + x_2^2 + \cdots + x_n^2)^{1/2}$

$d(x, y) = \|x - y\|$, $\rho(x, y) = \max\{|x_1 - y_1|, \cdots |x_n - y_n|\}$.

$\rho$ is called the square metric.

**Theorem:** The topologies induced by the euclidean metric $d$ and the square metric $\rho$ are the same as the product topology on $\mathbb{R}^n$.

**Proof:** Let $x, y \in \mathbb{R}^n$. Clearly $\rho(x, y) \leq d(x, y)$, for $B_d(x, \varepsilon) \subseteq B_\rho(x, \varepsilon)$.




$d(x, y) \leq \sqrt{n}\,\rho(x, y)$, so $B_\rho(x, \varepsilon/\sqrt{n}) \subseteq B_d(x, \varepsilon)$.
So both topologies are the same.

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So both topologies are the same.

Let $B = (a_1, b_1) \times \cdots \times (a_n, b_n)$ be a basis element of the product topology. For each $i$, there is a $\varepsilon_i$ such that $(x_i - \varepsilon_i, x_i + \varepsilon_i) \subseteq (a_i, b_i)$. Let $\varepsilon = \min\{\varepsilon_i\}$. Then $B_\rho(x, \varepsilon) \subseteq B$. Thus the $\rho$ topology is finer than the product topology.

Let $B_\rho(x, \varepsilon)$ be a basis element for the $\rho$ topology. Let $y \in B_\rho(x, \varepsilon)$. We must find $B$ basis element of the product topology such that $y \in B \subseteq B_\rho(x, \varepsilon)$. But

$$B_\rho(x, \varepsilon) = (x_1 - \varepsilon, x_1 + \varepsilon) \times \cdots \times (x_n - \varepsilon, x_n + \varepsilon)$$

is a basis element of the product topology. □

**Theorem:** Let $\bar{d}(a, b) = \min\{|a - b|, 1\}$ be the standard bounded metric on $\mathbb{R}$. If $x, y \in \mathbb{R}^\omega$, define

$$D(x, y) = \sup_i \left\{ \frac{\bar{d}(x_i, y_i)}{i} \right\}.$$

Then $D$ is a metric that induces the Product topology on $\mathbb{R}^\omega$.

**Proof:** Note that for all $i$, $\frac{\bar{d}(x_i, z_i)}{i} \leq \frac{\bar{d}(x_i, y_i)}{i} + \frac{\bar{d}(y_i, z_i)}{i}$

$$\leq D(x, y) + D(y, z).$$

So, $\sup_i \left\{ \frac{\bar{d}(x_i, z_i)}{i} \right\} = D(x, z) \leq D(x, y) + D(y, z)$.

If $D(x, y) = 0$, then each $\frac{\bar{d}(x_i, y_i)}{i} = 0 \Rightarrow x = y$.

each $\bar{d}$ is symmetric, so $D$ is as well.

Thus, $D$ is a metric on $\mathbb{R}^\omega$.

Let $U$ be open in the metric topology and let $x \in U$. Choose an $\varepsilon$-ball $B_D(x, \varepsilon)$ in $U$. Let $N$ s.t. $1/N < \varepsilon$. Let $V = (x_1 - \varepsilon, x_1 + \varepsilon) \times \cdots \times (x_N - \varepsilon, x_N + \varepsilon) \times \mathbb{R} \times \mathbb{R} \times \cdots$.

Let $y \in \mathbb{R}^\omega$. $\frac{\bar{d}(x_i, y_i)}{i} \leq 1/N$ for $i \geq N$. Thus

Let $y \in \mathbb{R}^\omega$. $\bar{d}(x_i, y_i) \leq 1/n$ for $i \geq n$. Thus

$$D(x,y) \leq \max\{\bar{d}(x_1,y_1), \ldots, \bar{d}(x_n,y_n), \tfrac{1}{n}\}.$$

If $y \in V$, then each $\bar{d}(x_i, y_i) < \varepsilon$, as is $1/n$, so $D(x,y) < \varepsilon$. Thus,

$$V \subseteq B_D(x, \varepsilon).$$

Thus the product topology is finer than the metric topology.

Let $U = \prod U_i$, where $U_i$ open in $\mathbb{R}$ for each $i = \alpha_1, \ldots, \alpha_n$ and $U_i = \mathbb{R}$ otherwise. Then $U$ is a basis element of the product topology. Let $x \in U$. We must find an open $V$ in the metric topology with $x \in V \subseteq U$. Choose an interval $(x_i - \varepsilon_i, x_i + \varepsilon_i)$ in $\mathbb{R}$ contained in $U_i$ for each $i \in \{\alpha_1, \ldots, \alpha_n\}$. Choose each $\varepsilon_i \leq 1$. Let

$$\varepsilon = \min\{\varepsilon_i / i \mid i \in \{\alpha_1, \ldots \alpha_n\}\}.$$

Let $y \in B_D(x, \varepsilon)$ Then for all $i$, $\bar{d}(x_i, y_i) \leq D(x,y) < \varepsilon$.

If $i \in \{\alpha_1, \ldots, \alpha_n\}$, then $\varepsilon \leq \varepsilon_i / i$, so

$$\bar{d}(x_i, y_i) < \varepsilon \Rightarrow \bar{d}(x_i, y_i) < i\varepsilon \leq \varepsilon_i \leq 1.$$

Thus, $|x_i - y_i| < \varepsilon_i$ so $y \in \prod U_i$.

□